

Regret-Based Portfolio Allocation: Minimizing Opportunity Cost and Its Dispersion

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ABSTRACT

Investors increasingly judge portfolios by how much they lag ex-post winners rather than by absolute Sharpe ratios, turning missed opportunities into a first-order driver of dissatisfaction. This paper addresses the resulting gap between regret-based evaluation and traditional risk models that manage the dispersion of portfolio returns instead of the dispersion of regret. We develop a Mean–MAD Regret framework that treats regret—not returns—as the primary object of risk management. The model defines period-by-period regret as the shortfall between a dynamic benchmark and portfolio returns, and then minimizes a Mean Absolute Deviation objective on this regret series using a single parameter to govern the trade-off between average regret and regret dispersion.

In a diversified multi-asset universe of equities, fixed income, commodities and real assets, regret-based portfolios with low to moderate dispersion aversion deliver higher compound returns and competitive or superior risk-adjusted performance relative to standard diversification heuristics and risk-based strategies, at the cost of greater concentration and drawdown risk. The main contribution is to provide an operational way to manage the level and volatility of opportunity cost relative to a dynamic benchmark, offering institutional investors a transparent tool to align portfolio construction with how performance is actually assessed.

TAKEAWAYS

- **Regret as the risk object, not returns.** The Mean–MAD Regret framework is the first to apply a dispersion measure (MAD) directly to the regret series—the opportunity cost relative to the ex-post best-performing asset—rather than to portfolio returns. This bridges the gap between how investors actually evaluate performance (relative to missed opportunities) and how portfolios are traditionally constructed.
- **A single parameter governs the entire concentration–diversification spectrum.** $\lambda = 0.1$ achieves a Sharpe ratio of 0.92 and CAGR of 14.82%, outperforming all nine benchmark strategies, while higher λ values progressively produce broader diversification. There is no single "optimal" λ : each institutional mandate maps naturally to a point on the spectrum, from aggressive growth to conservative pension allocations.
- **Outperformance is not an artifact of a single market history.** A Stationary Block Bootstrap with 1,000 replications and full re-optimization at every rebalancing date confirms that the CAGR advantage of low-dispersion-aversion specifications is robust to sampling variability, with non-overlapping 90% confidence intervals against five of the nine benchmarks. Bootstrap medians of MAD Regret decrease monotonically with λ , confirming that the dispersion-control mechanism operates systematically across synthetic return histories.

JEL: G11, G12, C61, D81

Keywords: Regret minimization, Mean Absolute Deviation, Portfolio optimization, Behavioral finance, Opportunity cost, Multi-asset allocation, Walk-forward backtesting, Stationary Block Bootstrap, Dynamic benchmark, Regret-diversification trade-off

1. INTRODUCTION

Classical portfolio allocation models, from Markowitz's (1952) mean–variance framework to modern Risk Parity approaches (Maillard et al., 2010), frame the investment decision as a trade-off between expected return and risk, typically measured through variance or related dispersion metrics of portfolio returns. In these settings, portfolios are evaluated in isolation: the objective is to shape a favorable absolute risk–return profile, with no explicit role for the psychological cost investors experience when comparing realized outcomes with alternatives that would have delivered higher performance.

A large body of evidence in behavioral finance shows that investors rarely assess performance in a vacuum. Both individuals and institutions routinely benchmark results against specific references—market indices, peer portfolios, or, in the extreme, the period's best-performing asset—amplifying hindsight bias and turning missed opportunities into a central driver of dissatisfaction (Bell, 1982; Loomes and Sugden, 1982). In such environments, investors effectively judge portfolios on regret—the opportunity cost relative to foregone alternatives—rather than on absolute metrics such as the Sharpe ratio, and well-diversified portfolios with positive absolute returns may still be perceived as failures relative to concentrated positions in ex-post winners. A growing theoretical literature formalizes this behavioral dimension, embedding regret into portfolio optimization through expected-regret minimization (Li et al., 2012), minimax-regret formulations (Edelman and Goryagina, 2019), retirement-specific regret models (Blanchett, 2023), and dynamic frameworks incorporating both regret aversion and rejoicing (Hambel and Dorelijers, 2025).

Separately, Mean Absolute Deviation (MAD) has emerged as a tractable alternative to variance for portfolio construction since the foundational work of Konno and Yamazaki (1991). Yet two limitations persist across these literatures. In regret-based models, regret typically enters as a scalar adjustment to expected utility or as a worst-case bound, without explicitly modelling its dispersion over time. In MAD and related risk-measure frameworks, dispersion control is applied to portfolio returns rather than to opportunity cost relative to a dynamic benchmark. This separation creates an integration gap that leaves an important dimension of investor concern—the volatility of regret itself—unmanaged. No existing framework allows an investor to explicitly trade off the level and the volatility of regret in a single convex optimization.

This paper addresses the gap by proposing a *Mean–MAD Regret framework* that applies a mean–MAD criterion directly to the time series of regret, defined as the shortfall between a dynamic benchmark process and the portfolio return. The framework treats the distribution of regret—not the distribution of returns—as the primary object of risk management, and introduces a single parameter λ to govern the trade-off between minimizing average regret and stabilizing regret dispersion. When $\lambda = 0$ the model reduces to pure expected-regret minimization; as λ increases, the optimizer increasingly

penalizes volatile opportunity cost, producing progressively more diversified allocations. The contribution is threefold: (i) methodologically, the paper demonstrates that applying MAD to regret yields a convex, tractable objective with desirable robustness properties; (ii) theoretically, it connects the criterion to the axiomatic regret literature, establishes a formal nesting relationship with classical tracking-error optimization, and positions the framework relative to online convex optimization; and (iii) empirically, it shows that regret-based portfolios with low to moderate dispersion aversion deliver competitive risk-adjusted performance in a diversified multi-asset universe, with robustness confirmed by a Stationary Block Bootstrap with full re-optimization.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature and provides axiomatic and decision-theoretic foundations for the criterion. Section 3 presents the mathematical framework. Section 4 describes the data and empirical design. Section 5 reports the main results and robustness checks. Section 6 concludes.

2. LITERATURE REVIEW

2.1 Regret theory and regret-based portfolio optimization

Regret theory, introduced independently by Bell (1982) and Loomes and Sugden (1982), models the emotional cost of comparing realized outcomes with foregone alternatives, leading to preferences that depend on both final wealth and counterfactual payoffs. Subsequent work establishes that regret-averse investors optimally internalize benchmark-relative underperformance and exhibit benchmark-tracking behavior, both in defined-contribution plans, where regret aversion helps explain demand for guarantees and balanced mixes (Muermann et al., 2006), and in institutional mandates, where explicit benchmark evaluation induces closer tracking and mitigates extreme positions (Wagner, 2002; Ibhagui, 2017).

Within portfolio optimization, regret has been embedded through two main channels. Expected-regret models define regret as the shortfall relative to an ex-post optimal or benchmark portfolio and show that minimizing expected regret typically leads to more diversified allocations than mean–variance, as diversification reduces the probability of extreme underperformance (Gao and Duan, 2010; Li et al., 2012; Liu, 2025). Minimax-regret approaches instead focus on worst-case shortfall relative to hindsight benchmarks and have been applied to pension and robust allocation problems, delivering lower downside risk and more stable funding ratios at the cost of some upside (Edelman and Goryagina, 2019; Caçador et al., 2022; Xidonas et al., 2017). A common feature of both channels, including "Markowitz with Regret" (Baule et al., 2018), is that regret enters as a scalar adjustment to expected utility or mean–variance criteria, so that the optimized object remains the distribution of portfolio returns. Even recent contributions

that combine regret with volatility risk measures treat regret as an auxiliary criterion rather than as the quantity whose dispersion is directly managed (Larni-Fooeik et al., 2024).

2.2 Mean Absolute Deviation and coherent risk measures

On the risk-measurement side, Mean Absolute Deviation has long served as an alternative to variance. Konno and Yamazaki (1991) introduce a mean–MAD model formulated as a linear program, and subsequent contributions extend it to skewness preferences, cardinality and transaction-cost constraints, robust estimation, and fuzzy representations (Konno and Suzuki, 1995; Konno and Wijayanayake, 1999; Moon and Yao, 2011; Deng and Geng, 2023). Empirical comparisons generally find that mean–MAD can match or improve on mean–variance risk–return trade-offs, sometimes yielding more stable weights and better out-of-sample performance under estimation error (DeMiguel et al., 2009; Martins, 2021). In parallel, coherent and convex risk measures such as CVaR (Artzner et al., 1999; Rockafellar and Uryasev, 2000) have been integrated with MAD and regret objectives in multi-objective and robust settings (Silva et al., 2017; Ararat et al., 2024), but the risk measure is still applied to portfolio returns, not to the regret series.

The result is an integration gap between two mature literatures. Regret-based models optimize a scalar regret quantity—its expectation or worst-case realization—without explicitly controlling how regret fluctuates across periods. MAD and CVaR frameworks provide precisely this kind of distributional control, but direct it at absolute returns rather than at opportunity cost. The Mean–MAD Regret framework closes this gap by applying a mean–MAD criterion directly to the time series of regret, making the distribution of opportunity cost the primary object of risk management.

2.3 Decision-theoretic foundations

The axiomatic regret literature provides a decision-theoretic rationale for treating regret as a first-order object. Savage's (1951) minimax regret criterion, axiomatized by Milnor (1954) and extended by Stoye (2011) to accommodate both prior-free and endogenous-prior formulations, focuses on worst-case regret. Sugden (1993) and Diecidue and Somasundaram (2017) provide axiomatic foundations for generalized regret theory, while Sarver (2008) axiomatizes preferences over menus in which agents anticipate experiencing regret when their choice turns out inferior to another available option, decomposing menu valuation into expected utility minus an anticipated regret penalty.

The key insight for the present framework is that Sarver's representation implies a clear separation between the utility of an alternative and the regret cost of comparison with foregone options. In the canonical regret-modified utility framework (Bell, 1982; Loomes and Sugden, 1982), an investor evaluating portfolio w relative to a benchmark process B_t derives per-period modified utility,

$$v(\mathbf{w}, t) = u\left(r_{p,t}(\mathbf{w})\right) - Q(\mathcal{R}_t(\mathbf{w})),$$

where $\mathcal{R}_t(\mathbf{w}) = B_t - r_{p,t}(\mathbf{w})$ denotes the period- t regret (formally developed in Section 3.2), and $Q(\cdot)$ is a convex, increasing regret function with $Q(0) = 0$, consistent with parametric forms studied by Bleichrodt and Wakker (2015) and the empirical measurements of Bleichrodt et al. (2010). When Q is approximately linear—a specification consistent with loss-aversion evidence—the ex-ante evaluation reduces to a function of two summary statistics of the regret distribution: its expected value and some measure of its dispersion $D(\cdot)$, yielding a functional of the form

$$V(\mathbf{w}) = -\mu\mathcal{R}(\mathbf{w}) - \lambda D(\mathcal{R}(\mathbf{w})),$$

directly analogous to mean–risk preferences over returns (Markowitz, 1952; Konno and Yamazaki, 1991) but with the risk measure applied to regret. Setting $D = \text{MAD}$ produces the Mean–MAD Regret criterion, with λ encoding aversion to regret volatility conditional on expected regret.

This formulation connects to a spectrum of existing regret decision rules. Savage's (1951) minimax regret corresponds to the limiting case $\lambda \rightarrow \infty$; expected-regret minimization (Li et al., 2012) corresponds to $\lambda = 0$. The Mean–MAD Regret framework generalizes both endpoints by allowing intermediate preferences over the regret distribution. From a risk-measurement perspective, MAD applied to the regret distribution inherits the robustness and tractability of the return-based case, and the composite functional is convex in w for any fixed benchmark, so any local minimum is a global minimum, avoiding spurious local optima and supporting stable numerical solutions.

2.4 Connections to tracking error and online learning

The Mean–MAD Regret framework is related to, but conceptually distinct from, tracking-error optimization (Roll, 1992; Jorion, 2003; Grinold and Kahn, 1999). Three differences are important. First, tracking error is symmetric—the manager can over- or under-perform a fixed, investable benchmark—whereas regret relative to the ex-post best-performing asset is inherently one-sided, capturing the opportunity cost of not having concentrated fully in the winner. Second, when the benchmark B_t is replaced by a fixed investable portfolio, regret reduces to the negative of excess return, and the Mean–MAD Regret problem becomes formally equivalent to a MAD-based tracking-error problem, nesting Roll's (1992) and Jorion's (2003) setting with MAD replacing variance. Third, the key innovation is the use of a non-investable, time-varying benchmark such as the perfect-foresight winner, which breaks the equivalence with standard tracking-error models and produces portfolios tilted toward assets with high recent winner frequencies, capturing a dimension of investment concern—the cost of missed opportunities—orthogonal to symmetric risk-budgeting.

The concept of regret relative to the best action in hindsight also has a well-developed formalization in online convex optimization (OCO) and universal portfolio theory. Cover (1991) introduces the universal portfolio, which achieves asymptotically zero regret per period relative to the best constant-rebalanced portfolio (CRP), and Cover and Ordentlich (1998) establish that worst-case regret against the best CRP is bounded by $O(N \log T / T)$. Helmbold et al. (1998) extend these results with multiplicative-update algorithms, while Hazan (2016) systematizes the OCO framework, showing that online gradient descent achieves $O(\sqrt{T})$ regret for convex losses. The present paper differs from this tradition in three respects: it is a statistical model minimizing expected regret and its dispersion over the empirical distribution rather than targeting worst-case bounds; the benchmark (the period-by-period best single asset) is more aggressive than the best CRP; and it explicitly controls the dispersion of regret through λ , a dimension absent from standard OCO formulations.

3. METHODOLOGY

3.1 Notation and Dynamic Benchmark

Consider an investment universe of N risky assets with simple returns $r_{i,t}$ for asset $i = 1, \dots, N$ in period $t = 1, \dots, T$. Let $\mathbf{r}_t = (r_{1,t}, \dots, r_{N,t})^\top$ denote the return vector. Portfolio weights at the beginning of period t are $\mathbf{w}_t = (w_{1,t}, \dots, w_{N,t})^\top$, subject to $\sum_{i=1}^N w_{i,t} = 1$ and $0 \leq w_{i,t} \leq \bar{w}_i$.

Weights are recalibrated only at predetermined rebalancing dates $\{\tau_1, \dots, \tau_K\}$, and held as a buy-and-hold portfolio between rebalancings. If $\mathbf{w}_\tau$ denotes the weight vector immediately after rebalancing, then for any $k \geq 1$ before the next rebalancing date τ' , the weight of asset i at $\tau + k$ is

$$w_{i,\tau+k} = \frac{w_{i,\tau} \prod_{s=\tau+1}^{\tau+k} (1 + r_{i,s})}{\sum_{j=1}^N w_{j,\tau} \prod_{s=\tau+1}^{\tau+k} (1 + r_{j,s})}, \quad i = 1, \dots, N$$

so that weights evolve with relative price movements while remaining fully invested. The portfolio return in period t is $r_{p,t}(\mathbf{w}) = \sum_{i=1}^N w_{i,t} r_{i,t}$. For notational convenience, R^p denotes the random portfolio-return variable with realizations $r_{p,t}(\mathbf{w})$; $\mu(R^p)$ and $\sigma(R^p)$ denote its mean and standard deviation, respectively.

To model regret flexibly, we introduce a *generic benchmark process* B_t representing the period- t return of the reference against which performance is evaluated. Benchmark choices include a market index, a liability-hedging portfolio, or a hindsight portfolio such as the best constant-mix benchmark used by Edelman and Goryagina (2019). In the empirical analysis, the baseline benchmark is $B_t = \max_{1 \leq i \leq N} r_{i,t}$, the return of the perfect-foresight best-performing asset each period. When the benchmark B_t is replaced

by a fixed investable portfolio, regret reduces to the negative of excess return, and the Mean–MAD Regret problem becomes formally equivalent to a MAD-based tracking-error problem, nesting Roll’s (1992) and Jorion’s (2003) settings with MAD replacing variance. In that case, the trivial zero-regret solution is to replicate the benchmark itself, so the main value of the present framework arises when regret is measured against non-investable or aspirational references such as the perfect-foresight winner.

3.2 Regret Definition and Risk Measures

Given a benchmark return B_t , period- t regret for a portfolio $\mathbf{w}$ is

$$\mathcal{R}_t(\mathbf{w}) = B_t - r_{p,t}(\mathbf{w})$$

This quantity captures the opportunity cost of not having invested according to the benchmark in period t . Under the perfect-foresight winner benchmark, $\mathcal{R}_t(\mathbf{w}) \geq 0$ for all feasible portfolios. From a behavioral perspective, this mirrors the counterfactual comparison central to regret theory (Bell, 1982; Loomes and Sugden, 1982; Wagner, 2002). The sequence $\{\mathcal{R}_t(\mathbf{w})\}_{t=1}^T$ defines a *regret distribution* that depends on $\mathbf{w}$ and on B_t ; in this paper, this regret distribution—not the return distribution—is the primary object of risk management.

Two statistics summarize the regret distribution: expected regret, $\mu_{\mathcal{R}}(\mathbf{w}) = \mathbb{E}[\mathcal{R}_t(\mathbf{w})]$, measuring the average opportunity cost; and the Mean Absolute Deviation of regret, $\text{MAD}_{\mathcal{R}}(\mathbf{w}) = \mathbb{E}[|\mathcal{R}_t(\mathbf{w}) - \mu_{\mathcal{R}}(\mathbf{w})|]$, capturing the typical magnitude of deviations of regret around its mean. The choice of MAD follows the mean–MAD portfolio literature (Konno and Yamazaki, 1991) and its extensions: MAD is more robust to outliers than variance, is homogeneous of degree one, and leads to convex optimization formulations. The critical distinction is that MAD is applied here to the regret distribution rather than to returns, shifting the focus from absolute risk to regret dispersion.

In practice, $\mu_{\mathcal{R}}$ and $\text{MAD}_{\mathcal{R}}$ are estimated from a rolling historical window of length T_{train} at each rebalancing date τ :

$$\hat{\mu}_{\mathcal{R}}(\mathbf{w}; \tau) = \frac{1}{T_{\text{train}}} \sum_{t=\tau-T_{\text{train}}}^{\tau-1} \mathcal{R}_t(\mathbf{w}), \quad \widehat{\text{MAD}}_{\mathcal{R}}(\mathbf{w}; \tau) = \frac{1}{T_{\text{train}}} \sum_{t=\tau-T_{\text{train}}}^{\tau-1} |\mathcal{R}_t(\mathbf{w}) - \hat{\mu}_{\mathcal{R}}(\mathbf{w}; \tau)|,$$

ensuring that only information available at time τ is used. Alternative dispersion measures—such as the standard deviation, semi-variance, or CVaR of regret—can be embedded in the same framework, linking to the broader literature on coherent risk measures (Artzner et al., 1999; Rockafellar and Uryasev, 2000). In this paper, we focus on MAD of regret for tractability and robustness, leaving a systematic comparison with tail-focused regret measures to future work.

3.3 Objective Function

The Mean–MAD Regret portfolio at rebalancing date τ solves

$$\min_{\mathbf{w}} \hat{\mu}_{\mathcal{R}}(\mathbf{w}; \tau) + \lambda \widehat{\text{MAD}}_{\mathcal{R}}(\mathbf{w}; \tau)$$

subject to $\sum_{i=1}^N w_i = 1$, $0 \leq w_i \leq \bar{w}_i$, where $\lambda \geq 0$ controls the weight placed on regret dispersion relative to average regret. This is directly analogous to the classical mean–MAD framework (Konno and Yamazaki, 1991) but applied to the regret distribution rather than to returns. In what follows, we refer to this allocation rule as the Mean–MAD Regret (MMR) framework and denote its portfolios as $\text{MMR}(\lambda)$. When $\lambda = 0$, the problem reduces to pure expected regret minimization in the spirit of Li et al. (2012), favoring concentrated portfolios in assets that frequently outperform the benchmark. While the optimization is numerically well behaved, the resulting allocations are economically fragile, exhibiting low diversification and exposure to deep drawdowns. As λ increases, the optimization increasingly penalizes volatile regret, promoting more stable benchmark-relative performance even at the cost of somewhat higher average regret. For sufficiently large λ , solutions converge toward highly diversified but generally not equal-weight allocations, as assets that rarely rank among the top performers receive persistently low weight even when dispersion minimization dominates. The decision-theoretic motivation for this trade-off is developed in Section 2.3.

The parameter λ represents an investor preference—analogueous to the risk-aversion coefficient in mean–variance optimization—rather than a statistical quantity to be estimated from data. Different mandates, governance structures, and regulatory constraints imply heterogeneous tolerances for regret level and volatility, so the framework does not aim to identify a single “optimal” λ . Section 4.2 describes the representative grid used in the empirical analysis.

Since $\mathcal{R}_t(\mathbf{w})$ is affine in $\mathbf{w}$ for any fixed benchmark B_t , each term $|\mathcal{R}_t(\mathbf{w}) - \hat{\mu}_{\mathcal{R}}|$ is convex as the absolute value of an affine function, and a non-negative linear combination of convex functions is convex. Combined with the convexity of the feasible set, this guarantees that any local minimum is a global minimum, avoiding spurious local optima and supporting stable numerical solutions.

3.4 Implementation

Numerical optimization. Although the Mean–MAD Regret objective can be reformulated as a linear program by introducing auxiliary variables for absolute deviations, it is solved with Sequential Least Squares Programming (SLSQP; Kraft, 1988), which handles the budget and box constraints efficiently. At each rebalancing date, a multi-start scheme with three initializations—Equal Weight, Winner Frequency, and a Dirichlet draw ($\alpha = 1$)—is used to mitigate sensitivity to starting values, with the

solution yielding the lowest objective value selected. Convergence tolerance is set to 10^{-9} with a maximum of 500 iterations; the solver converges within the iteration limit for all dates and strategies. All benchmark strategies (mean–variance, risk-based, hierarchical, and diversification-based) are implemented with the same SLSQP solver (or closed-form rules where available) under identical constraints, so that performance differences reflect the allocation rules themselves rather than solver choice.

Transaction costs and NAV evolution. Once optimal weights $\mathbf{w}_\tau^{\text{opt}}$ are obtained, they are implemented at the beginning of the next period and held until the following rebalancing, with weights evolving according to buy-and-hold drift. Transaction costs are incorporated at each rebalancing. Let $\mathbf{w}_\tau^{\text{pre}}$ denote the pre-rebalancing weights; one-way turnover is $\text{TO}_\tau = \frac{1}{2} \sum_{i=1}^N |w_{i,\tau}^{\text{opt}} - w_{i,\tau}^{\text{pre}}|$, and the cost impact is $\text{Cost}_\tau = c \cdot \text{TO}_\tau$, where c is the round-trip proportional cost rate. The NAV evolves as

$$\text{NAV}_t = \text{NAV}_{t-1} \left(1 + r_{p,t}(\mathbf{w}) \right) (1 - \text{Cost}_t \cdot \mathbb{1}_{t \in \{\tau_k\}}).$$

3.5 Out-of-Sample Validation

Walk-forward rolling-window design

The empirical analysis uses a walk-forward rolling-window protocol in which, at each rebalancing date, a fixed-length training window is used to estimate all model inputs (regret moments for MMR; returns, covariances and volatilities for benchmarks), after which optimized weights are implemented over the subsequent out-of-sample period. Training-window length, rebalancing frequency, constraint set, and transaction-cost assumptions are held constant across all strategies, ensuring a like-for-like comparison: differences in performance reflect the allocation rules themselves rather than differences in information sets or implementation conditions.

Stationary block bootstrap with re-optimization

To assess robustness to sampling variability and path dependence, a Stationary Block Bootstrap (SBB; Politis and Romano, 1994; Künsch, 1989) is applied to the full panel of asset returns, and the entire walk-forward procedure is re-run within each bootstrap sample. Unlike simple bootstraps that resample only final portfolio returns, this approach resamples the underlying return history, re-estimates model inputs and re-solves each optimization problem, so that parameter-estimation risk, optimization sensitivity and path dependence are all reflected in the bootstrap distributions. The SBB generates blocks of geometrically distributed length with mean l , preserving temporal dependence within blocks while randomizing their order across time. The average block length is set equal to the training window ($l = 156$ weeks ≈ 3 years), so that each bootstrap block preserves the full dependence structure over which model inputs are estimated. This choice ensures that within-block serial correlation, momentum persistence, and regime

transitions that span the estimation horizon are retained in the resampled paths, rather than being fragmented by shorter blocks. While shorter block lengths (e.g., $l = 12$ –26 weeks) are common in asset-pricing tests where the object of interest is a single return series, the present application re-optimizes portfolios at each rebalancing date using a rolling training window of identical length; matching l to this window guarantees that each synthetic training sample is internally coherent, preventing artificial breaks in the dependence structure that the optimizer exploits.

Applying the algorithm to the multivariate return series preserves cross-sectional dependence at each date. For each of B bootstrap replications, empirical distributions of performance and risk metrics, along with regret level and dispersion, are constructed; percentile-based confidence intervals across strategies provide evidence on whether observed performance differences are robust to alternative market histories.

4. DATA AND BENCHMARKS

4.1 Asset Universe and Data

The empirical analysis uses a diversified multi-asset universe of fourteen highly liquid U.S.-listed exchange-traded funds (ETFs) spanning equities, fixed income, real estate and commodities, as summarized in Table 1. Weekly adjusted close prices are obtained from Bloomberg for April 2007 through December 2025 and converted to simple returns. The sample covers multiple market regimes, including the global financial crisis (2008–2009), the post-crisis bull market (2010–2019), the COVID-19 shock (2020) and the inflation/monetary-tightening period (2022–2023).

TABLE 1. *Multi-asset ETF universe*

Ticker	Asset Class	Description
SPY	U.S. Equity	SPDR S&P 500 ETF Trust
QQQ	U.S. Equity (Growth)	Invesco QQQ Trust (Nasdaq-100)
IWM	U.S. Equity (Small Cap)	iShares Russell 2000 ETF
EFA	International Equity	iShares MSCI EAFE ETF (Developed ex-U.S.)
EEM	Emerging Markets Equity	iShares MSCI Emerging Markets ETF
IGOV	International Fixed Income	iShares International Treasury Bond ETF
AGG	US Fixed Income	iShares Core U.S. Aggregate Bond ETF
LQD	Corporate Credit	iShares iBoxx \$ Investment Grade Corporate Bond ETF
HYG	High Yield	iShares iBoxx \$ High Yield Corporate Bond ETF
VNQ	US Real Estate	Vanguard Real Estate ETF (U.S. REITs)
RWX	International Real Estate	SPDR Dow Jones International Real Estate ETF
GLD	Commodities (Gold)	SPDR Gold Trust (Physical Gold)
DBC	Commodities (Broad)	Invesco DB Commodity Index Tracking Fund
TIP	US Inflation-Protected Bond	iShares TIPS Bond ETF

The universe is designed to provide broad coverage across asset classes: the equity sleeve includes U.S. large-cap, growth, small-cap, developed ex-U.S. and emerging-markets

exposures; the fixed-income segment comprises U.S. aggregate bonds, international bonds, inflation-linked bonds, investment-grade credit and high yield; and real assets are represented by U.S. REITs and international real estate, and commodity exposure via gold and a broad commodity index. All ETFs are highly traded, with long-lived histories and narrow bid-ask spreads, making them suitable for the implementation of dynamic allocation strategies with modest transaction-cost assumptions. Transaction costs are set to 30 basis points per round trip. All strategies use identical training windows, rebalancing frequency and transaction-cost assumptions. Table 2 summarizes the empirical parameter choices.

TABLE 2: *Empirical Parameter Choices*

Parameter	Specification	Rationale
Periodicity	Weekly data	Balance information content and microstructure noise
Training window	156 weeks (≈ 3 years)	Sufficient history for stable estimation; standard in tactical allocation research
Rebalancing frequency	26 weeks (semi-annual)	Reflects strategic allocation; reduces turnover and costs
Out-of-sample period	June 2010 – December 2025	15.5 years, 31 rebalancings; multiple market regimes
Reserve period	May 2007 – May 2010	Initial estimation only; ensures strict out-of-sample evaluation
Transaction costs	30 bps round-trip	Consistent with liquid ETF bid-ask spreads and commissions
Optimization algorithm	SLSQP with multi-start	Three initializations (EW, Winner Freq, Dirichlet $\alpha=1$); tolerance = 10^{-9}
Bootstrap parameters	$B = 1,000$, block length $l = 156$ weeks (≈ 3 years)	Stationary Block Bootstrap; block length matches the training window, preserving full estimation-cycle dependence

4.2 Benchmark Strategies

The MMR portfolios are constructed relative to a perfect-foresight benchmark that allocates 100% of capital in each period to the ex-post best-performing asset. This benchmark does not imply informational advantages for the optimizer: the identity of the best asset is unknown ex ante, and the benchmark serves as a theoretical upper bound that induces a natural diversification incentive. While aggressive, this benchmark is intentionally aspirational and not meant to represent a feasible investment strategy. This definition is standard in the regret-minimizing and universal portfolio literatures (Cover, 1991; Cover and Ordentlich, 1998; Helmbold et al., 1998). Unreported robustness checks using softer top- K benchmarks—defined as the average return of the K best-performing assets each period, with $K = 2$ and $K = 3$ —confirm that the regret-diversification trade-off and the qualitative ranking across λ specifications are preserved under less extreme reference points.

The framework evaluates MMR portfolios across four representative values $\lambda \in \{0, 0.1, 1, 10\}$ spanning an economically meaningful range: $\lambda = 0$ (pure expected-regret minimization, aggressive), $\lambda = 0.1$ (low dispersion penalty, mildly regret-averse), $\lambda = 1$ (balanced trade-off), and $\lambda = 10$ (strong dispersion aversion, conservative). These values are chosen to cover the spectrum from highly concentrated to highly diversified solutions without imposing a single “optimal” choice.

The regret-based portfolios are further compared against nine benchmark strategies encompassing passive allocations, risk-based portfolios, mean–variance optimization, hierarchical methods and diversification-oriented approaches, as summarized in Table 3.

TABLE 3: *Benchmark Portfolio Strategies*

Category	Strategy	Optimization Criterion / Weighting Rule	Reference
Passive/Heuristic	Equal Weight (EW)	$w_i = 1/N$	DeMiguel et al. (2009)
	Winner Frequency (WF)	$w_i \propto \text{Freq (asset } i = \text{best performer)}$	–
Risk-Based	Risk Parity (RP): Equal Risk Contribution	$w_i \frac{\partial \sigma(R^p)}{\partial w_i} = \text{constant}$	Maillard et al. (2010); Roncalli (2013)
	Inverse Volatility (IV)	$w_i \propto 1/\sigma_i$	–
Mean-Variance	Minimum Variance (MVO)	$\min_{\mathbf{w}} \sigma^2(R^p)$	Markowitz (1952)
	Maximum Sharpe (MVO)	$\max_{\mathbf{w}} \frac{\mu(R^p) - r_f}{\sigma(R^p)}$	Markowitz (1952)
Hierarchical clustering	Hierarchical Risk Parity (HRP)	Hierarchical clustering + inverse-variance weighting	López de Prado (2016)
	Hierarchical Equal Risk Contribution (HERC)	Hierarchical clustering + equal risk contribution	Raffinot (2018)
Diversification	Diversification Ratio (DR)	$\max_{\mathbf{w}} \frac{\sum_i w_i \sigma_i}{\sigma(R^p)}$	Choueifaty & Coignard (2008)

Winner Frequency weights are set proportional to the empirical frequency with which each asset has been the best performer over the training window. Specifically, each period casts a single unweighted vote for the asset with the highest return; weights are then proportional to vote counts, so the margin of outperformance does not affect the allocation. Risk-based strategies allocate capital according to risk contributions or inverse volatilities (Maillard et al., 2010; Roncalli, 2013), hierarchical methods exploit clustering to distribute risk across groups (López de Prado, 2016; Raffinot, 2018), and Diversification Ratio maximizes the ratio of weighted individual volatilities to portfolio volatility (Choueifaty and Coignard, 2008).

5. EMPIRICAL RESULTS

5.1 Performance Analysis

Table 4 summarizes out-of-sample performance for the four MMR portfolios and nine benchmark strategies over the evaluation period June 2010–December 2025. The Probabilistic Sharpe Ratio (*PSR*) of Bailey and López de Prado (2012), which tests whether each strategy’s observed Sharpe ratio is statistically distinguishable from zero after adjusting for skewness, kurtosis and sample length, is reported alongside the point estimate. The last two columns report expected regret and its mean absolute deviation, providing a common opportunity-cost metric for all strategies regardless of their construction principle.

TABLE 4. *Out-of-Sample Performance (June 2010 – December 2025)*

Strategy	CAGR (%)	Ann Vol (%)	SR	PSR (SR*=0)	MaxDD (%)	Entropy $\eta(w)$	E[Regret]	MAD[Regret]
MMR(λ)								
$\lambda = 0$ (Pure Regret)	16.10	17.79	0.93	1.000	−37.63	1.03	2.32	1.72
$\lambda = 0.1$ (Low dispersion aversion)	14.82	16.60	0.92	1.000	−31.68	1.48	2.35	1.51
$\lambda = 1$ (Balanced)	8.10	12.66	0.68	0.996	−24.56	5.45	2.47	1.06
$\lambda = 10$ (High dispersion aversion)	5.56	11.26	0.54	0.981	−22.91	7.05	2.52	1.00
HEURISTIC								
Equal Weight	4.86	10.28	0.51	0.976	−23.21	14.00	2.54	0.98
Winner Frequency	6.91	12.36	0.60	0.990	−27.08	9.43	2.49	0.97
RISK-BASED								
Risk Parity (ERC)	3.00	6.71	0.47	0.967	−19.48	10.78	2.58	1.12
Inverse Volatility	2.95	7.10	0.44	0.958	−20.25	11.49	2.58	1.10
MEAN-VARIANCE								
Minimum Variance	0.74	4.65	0.18	0.763	−19.98	3.82	2.62	1.35
Maximum Sharpe	9.55	11.88	0.83	0.999	−25.67	2.75	2.44	1.30
HIERARCHICAL								
HRP	1.17	5.41	0.24	0.831	−19.14	7.35	2.61	1.27
HERC	0.81	4.96	0.19	0.772	−20.10	5.97	2.62	1.34
DIVERSIFICATION								
Diversification Ratio	3.83	6.21	0.64	0.993	−17.22	5.04	2.56	1.16

Notes: All strategies employ identical walk-forward specifications (156-week training window, semi-annual rebalancing, 30 bps round-trip transaction costs). Sharpe ratios (Sharpe, 1966, 1994) use excess returns over a zero risk-free rate, following standard practice in multi-asset backtesting studies that span heterogeneous asset classes where no single risk-free rate is unambiguously applicable (Pezier and White, 2008). PSR = Probabilistic Sharpe Ratio with benchmark $SR^* = 0$ (Bailey and López de Prado, 2012), offering a neutral reference that makes *PSR* comparable across all strategies. Entropy = mean per-rebalancing exponential of Shannon entropy (Meucci, 2009), measuring the average effective number of assets held. E[Regret] = mean weekly shortfall relative to the ex-post best-performing asset. MAD[Regret] = mean absolute deviation of regret.

Figures 1 and 2 show the out-of-sample evolution of portfolio weights for the nine benchmark strategies and the regret-based portfolios.

FIGURE 1. *Portfolio Strategies: Out-of-sample weights*

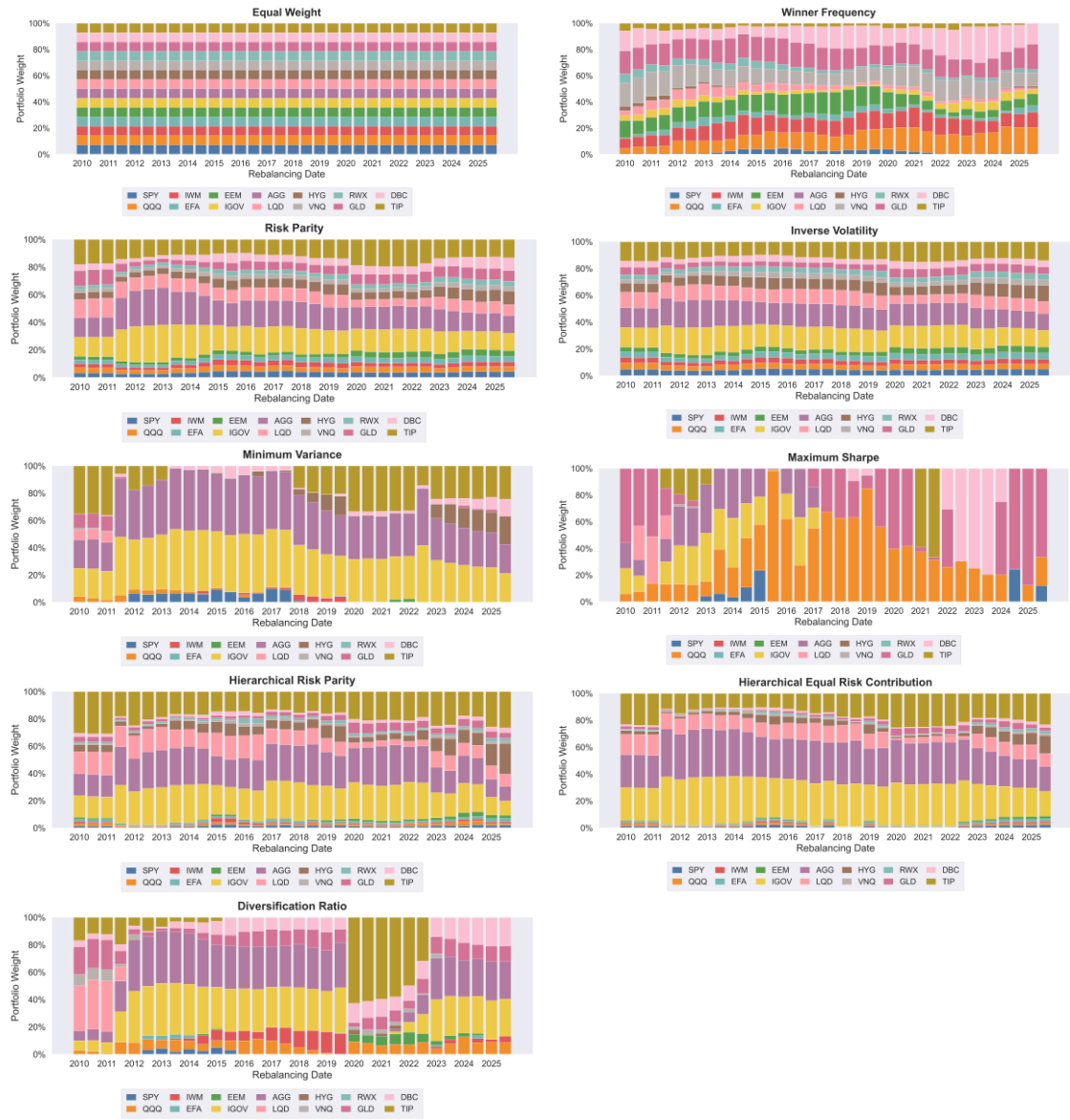
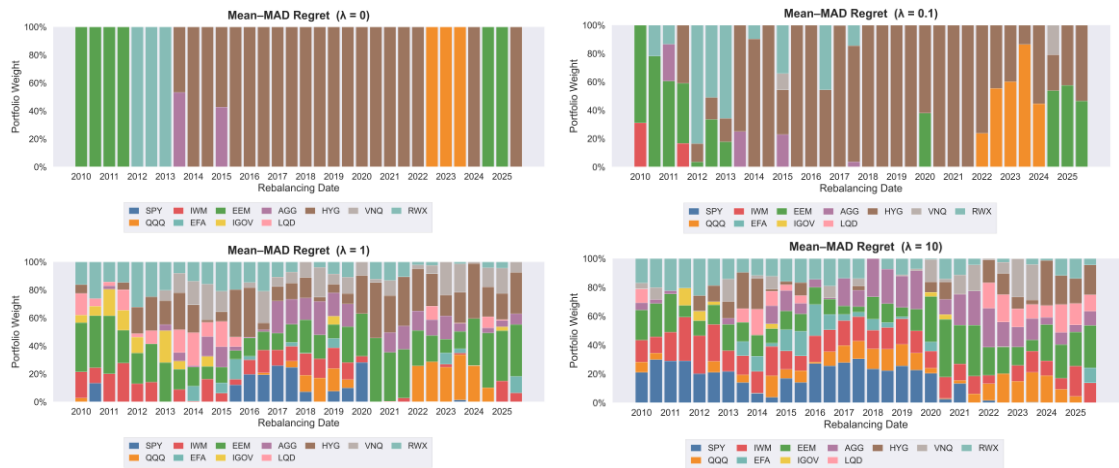


Figure 2. *Regret Portfolio: Out-of-sample weights per different lambda values*



Regret–diversification trade-off

The four rows at the top of Table 4 show how varying λ affects performance, diversification and regret. Pure expected-regret minimization ($\lambda = 0$) delivers the highest CAGR (16.10%) and Sharpe ratio (0.93), with $PSR = 1.000$. This comes at a clear cost: annualized volatility of 17.79%, a maximum drawdown of -37.63% , and only 1.0 effective asset — essentially a single-asset bet. The combination of high CAGR and near-unit entropy for $\lambda = 0$ illustrates the core trade-off: expected-regret minimization delivers strong headline performance but produces portfolios that are closer to “winner-take-all” bets than to institutionally robust allocations. The regret profile mirrors this pattern: $E[\text{Regret}]$ is the lowest in the sample (2.32%), but $MAD[\text{Regret}]$ is the highest (1.72%), pointing to large swings in weekly opportunity cost.

Introducing a small dispersion penalty ($\lambda = 0.1$) preserves most of the return advantage (CAGR 14.82%, Sharpe 0.92, PSR 1.000) while materially reducing maximum drawdown from -37.63% to -31.68% and $MAD[\text{Regret}]$ from 1.72% to 1.51% at virtually no cost in expected regret. The balanced specification ($\lambda = 1$) shifts toward broader diversification: entropy rises to 5.4 effective assets, maximum drawdown narrows to -24.56% , and $MAD[\text{Regret}]$ falls to 1.06%, while CAGR declines to 8.10% and PSR remains high at 0.996. High dispersion aversion ($\lambda = 10$) approaches an asymptotic diversification regime with approximately 7 effective assets and $MAD[\text{Regret}]$ at 1.00%, but CAGR drops to 5.56%. The monotonic progression confirms that λ provides effective and transparent control over the regret–diversification trade-off.

Comparison with benchmarks

Equal Weight delivers a CAGR of 4.86% (Sharpe 0.51, PSR 0.976); the CAGR gap to $\lambda = 0.1$ is nearly 10 percentage points, and the Sharpe difference (0.92 vs. 0.51) is economically substantial. Winner Frequency achieves a higher CAGR (6.91%, Sharpe 0.60, PSR 0.990), confirming that winner-based information is relevant, but falls short of the regret-based portfolios that optimize over this information. Risk Parity attains a Sharpe of 0.47 with the lowest maximum drawdown among broadly diversified strategies (-19.48%), but its CAGR of 3.00% illustrates the opportunity cost of risk equalization. Maximum Sharpe delivers the highest CAGR among non-regret strategies (9.55%, Sharpe 0.83, PSR 0.999), but falls below $\lambda = 0.1$ on both CAGR and Sharpe despite relying on explicit expected-return estimation, consistent with the well-documented sensitivity of mean–variance optimization to estimation error (Chopra and Ziemba, 1993; DeMiguel et al., 2009). Minimum Variance (CAGR 0.74%, PSR 0.763), HRP (1.17%, PSR 0.831), and HERC (0.81%, PSR 0.772) fail to achieve statistically significant Sharpe ratios at the 95% level. Diversification Ratio (Sharpe 0.64, PSR 0.993) achieves the shallowest maximum drawdown (-17.22%) but a CAGR of only 3.83%.

Three conclusions emerge. First, regret-based portfolios with low dispersion aversion ($\lambda \leq 0.1$) achieve the highest compound returns and risk-adjusted performance, with PSR of 1.000 for both specifications. Second, PSR reveals meaningful differences in statistical reliability: all regret specifications achieve PSR above 0.98, as do Maximum Sharpe and

Diversification Ratio, whereas Minimum Variance, HRP, and HERC fall below 0.95. Third, the parameter λ allows investors to navigate the full spectrum from concentrated, return-maximizing allocations to broadly diversified portfolios via a single, economically interpretable control.

5.2 Bootstrap Robustness

The walk-forward backtest operates on a single realized path of returns. To assess whether the documented outperformance is structural or sample-specific, a Stationary Block Bootstrap (SBB) with $B = 1,000$ replications and average block length $l = 156$ weeks is applied, with each replication re-running the complete walk-forward procedure with full portfolio re-optimization at every rebalancing date, so that estimation error, rebalancing timing, and transaction costs are all reflected in the simulated distributions. Table 5 reports 90% confidence intervals ([p5, p95]) for six key metrics. Figures 3 through 6 provide a visual complement to Table 5.

TABLE 5. *Bootstrap Robustness: 90% Confidence Intervals*

(Stationary Block Bootstrap, $B = 1,000$ replications, average block length $l = 156$ weeks, full re-optimization at each rebalancing date)

Strategy	CAGR (%)		Sharpe Ratio		Max Drawdown (%)		Entropy $\eta(w)$		Mean Regret (%)		MAD Regret (%)	
	Obs.	[p5, p95]	Obs.	[p5, p95]	Obs.	[p5, p95]	Obs.	[p5, p95]	Obs.	[p5, p95]	Obs.	[p5, p95]
Panel A — Mean-MAD Regret Framework												
Mean-MAD Regret ($\lambda = 0$)	16.10	[4.25, 20.62]	0.930	[0.301, 1.104]	-37.63	[-63.87, -27.25]	1.03	[1.00, 1.09]	2.32†	[2.22, 3.17]	1.72†	[1.61, 2.45]
Mean-MAD Regret ($\lambda = 0.1$)	14.82	[5.09, 19.41]	0.917	[0.368, 1.096]	-31.68	[-59.95, -24.59]	1.48	[1.31, 1.84]	2.35†	[2.24, 3.18]	1.51†	[1.43, 2.16]
Mean-MAD Regret ($\lambda = 1$)	8.10	[2.92, 12.39]	0.679	[0.280, 0.927]	-24.56	[-46.84, -20.85]	5.45	[4.42, 5.98]	2.47†	[2.39, 3.26]	1.06†	[1.00, 1.66]
Mean-MAD Regret ($\lambda = 10$)	5.56	[1.03, 9.30]	0.537	[0.148, 0.768]	-22.91	[-45.76, -21.38]	7.05	[6.21, 7.59]	2.52†	[2.43, 3.31]	1.00†	[0.95, 1.59]
Panel B — Benchmark Strategies												
Equal Weight	4.86	[0.55, 6.63]	0.513	[0.107, 0.633]	-23.21	[-46.92, -22.19]	14.00	[14.00, 14.00]	—	—	—	—
Risk Parity	3.00	[0.37, 4.48]	0.475	[0.087, 0.673]	-19.48	[-32.35, -14.45]	10.78	[10.15, 11.50]	—	—	—	—
Winner Frequency	6.91	[2.04, 9.69]	0.603	[0.214, 0.762]	-27.08	[-49.80, -20.24]	9.43	[8.91, 9.92]	—	—	—	—
Max Sharpe (MVO)	9.55	[3.39, 14.21]	0.828	[0.328, 1.126]	-25.67	[-50.11, -16.61]	2.75	[2.27, 3.44]	—	—	—	—
Min Variance (MVO)	0.74	[-1.28, 2.37]	0.181	[-0.211, 0.470]	-19.98	[-31.04, -9.01]	3.82	[3.43, 4.46]	—	—	—	—
HRP	1.17	[-0.74, 2.69]	0.243	[-0.093, 0.485]	-19.14	[-30.47, -11.50]	7.35	[6.88, 7.99]	—	—	—	—
Inv. Volatility	2.95	[0.09, 4.32]	0.445	[0.053, 0.604]	-20.25	[-34.93, -15.73]	11.49	[11.18, 11.89]	—	—	—	—
HERC	0.81	[-0.97, 2.44]	0.188	[-0.144, 0.465]	-20.10	[-32.93, -9.77]	5.97	[5.71, 7.20]	—	—	—	—
Diversification Ratio	3.83	[1.07, 5.64]	0.636	[0.202, 0.863]	-17.22	[-29.24, -11.89]	5.04	[4.81, 5.77]	—	—	—	—

Notes: “Obs.” denotes the point estimate from the original out-of-sample backtest. [p5, p95] reports the 5th and 95th percentile of the SBB distribution. Mean Regret and MAD Regret are reported for Mean-MAD Regret strategies only, as these are the quantities being optimized. † indicates the observed value lies within the bootstrap interval. Negative lower bounds for Sharpe and CAGR in some benchmarks reflect adverse synthetic paths.

FIGURE 3. *CAGR Bootstrap Confidence Intervals*

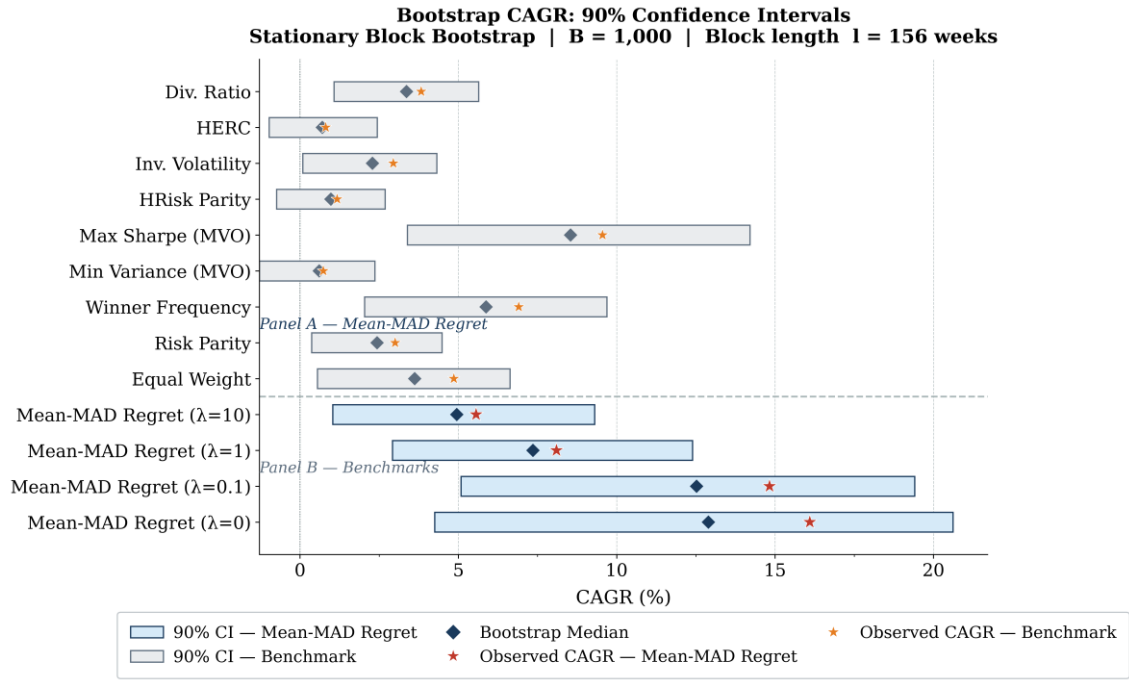


Figure 3 displays bootstrap 90% confidence intervals for CAGR across all strategies, confirming that $MMR(\lambda = 0.1)$ maintains a materially higher return distribution than most benchmarks even under resampled market histories.

FIGURE 4. *Sharpe Ratio Bootstrap Confidence Intervals*

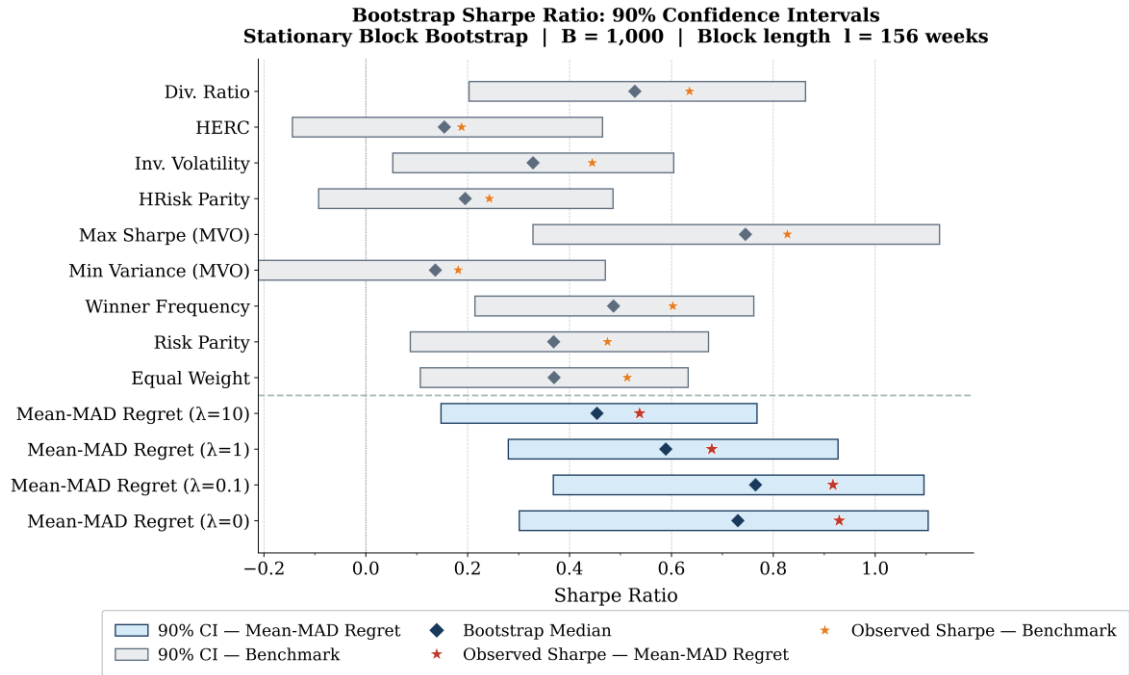


Figure 4 reports the corresponding intervals for the Sharpe ratio, where MMR specifications with low dispersion aversion occupy the upper region of the distribution alongside Maximum Sharpe, while several benchmarks include zero within their confidence bands.

FIGURE 5. Portfolio Entropy Bootstrap Confidence Intervals

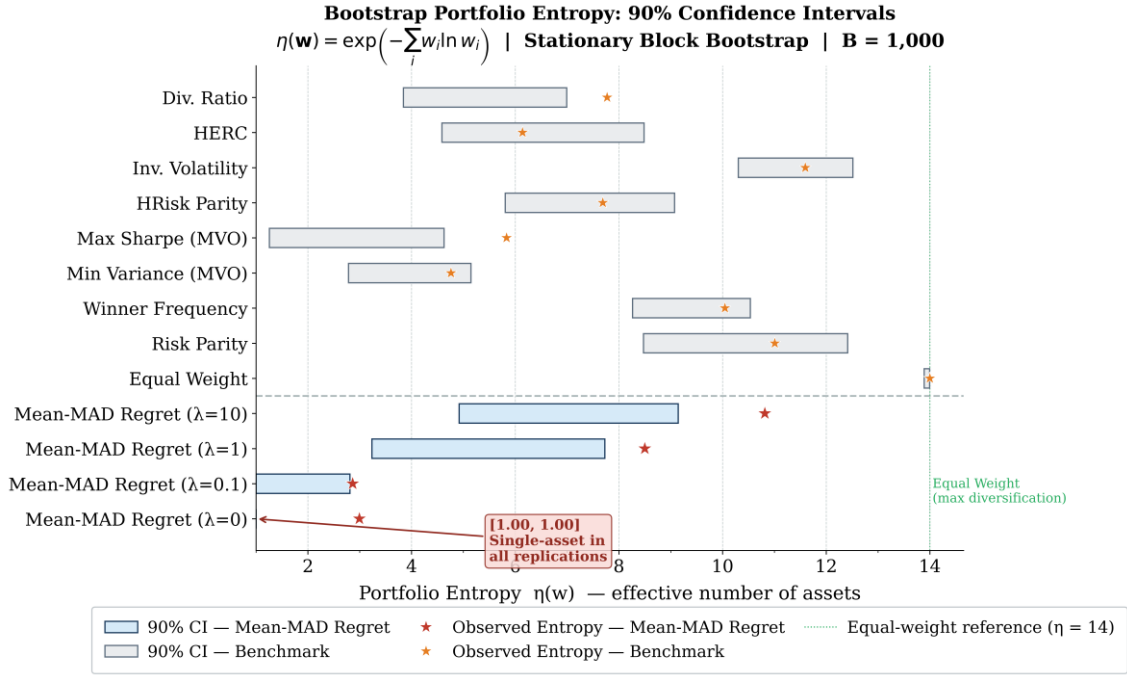


Figure 5 displays entropy intervals, confirming that the progression from concentrated to diversified allocations as λ increases is a structural property of the objective function rather than a sample-period artefact: bootstrap entropy for $\lambda = 0$ collapses to a single effective asset across virtually all replications, while $\lambda = 10$ consistently maintains six to eight effective assets.

FIGURE 6. MAD Regret Bootstrap Confidence Intervals by λ

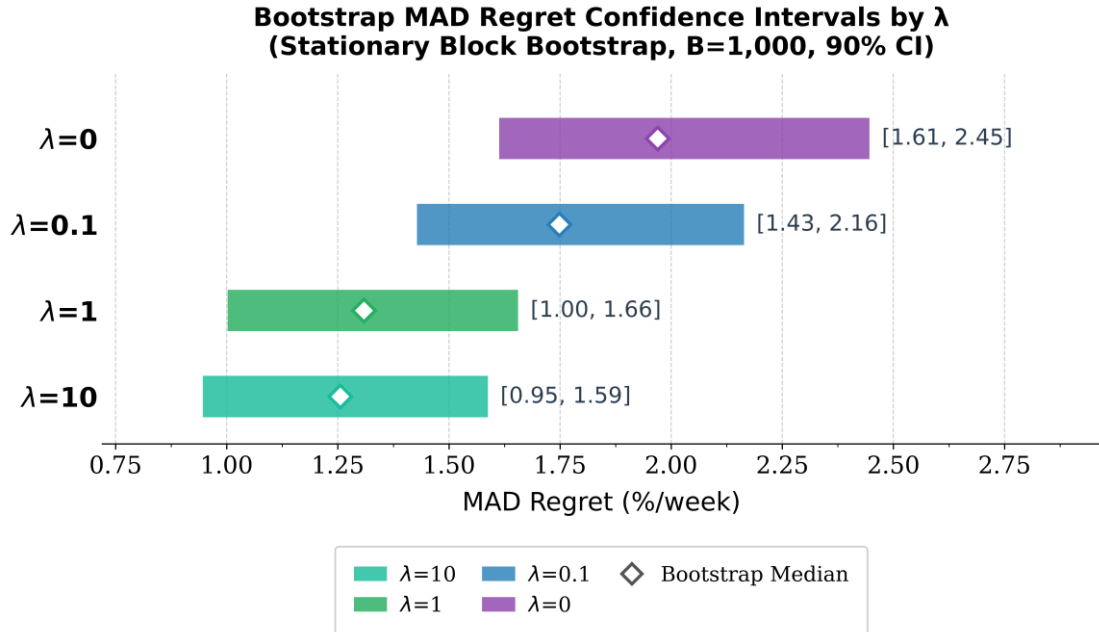


Figure 6 focuses exclusively on the regret dimension, plotting MAD Regret intervals for the four λ values; bootstrap medians decrease monotonically with λ , and the progressive leftward shift of the confidence intervals confirms that the dispersion-control mechanism operates systematically across synthetic return histories.

The bootstrap analysis validates the core claims along four dimensions. First, CAGR outperformance of low-dispersion-aversion specifications over most benchmarks is not a single-path artefact: across the 1,000 bootstrap replications, the p5 of $\text{MMR}(\lambda = 0.1)$ CAGR (5.09%) strictly exceeds the p95 of five out of the nine benchmark strategies (Risk Parity, Min Variance, HRP, Inv. Volatility, and HERC), establishing first-order stochastic dominance in the bootstrap distribution for this subset. Maximum Sharpe (p95 = 14.21%) and Winner Frequency (p95 = 9.69%) exhibit substantial overlap with $\text{MMR}(\lambda = 0.1)$, while Equal Weight (p95 = 6.63%) and Diversification Ratio (p95 = 5.64%) show only marginal overlap, with their p95 exceeding the $\text{MMR}(\lambda = 0.1)$ p5 by fewer than 2 percentage points.

Second, the monotone relationship between λ and realized MAD Regret is preserved across the bootstrap distribution: bootstrap medians decrease monotonically from 1.97% ($\lambda = 0$) to 1.26% ($\lambda = 10$), and the confidence intervals shift progressively leftward with increasing λ , confirming that the dispersion-reduction mechanism operates systematically under a wide range of synthetic return histories. Adjacent intervals partially overlap, indicating that the incremental effect of each λ step, while directionally consistent, is not individually significant at the 90% level.

Third, drawdown risk is structurally understated in concentrated strategies: for $\lambda = 0$, the observed Max Drawdown is -37.63% yet the bootstrap p5 reaches -63.87% , and for $\lambda = 1$ the observed drawdown of -24.56% has a bootstrap median exceeding -35% . By contrast, Risk Parity (p5 = -32.35%) and Diversification Ratio (p5 = -29.24%) show far more stable distributions. This reinforces the institutional necessity of a positive λ .

Fourth, entropy analysis confirms that single-asset convergence at $\lambda = 0$ is an inherent property of the unconstrained objective, not a sample-period coincidence: bootstrap p5 entropy for $\lambda = 0$ reaches 1.00 (single-asset portfolio) across all replications, while $\lambda = 1$ maintains 4–6 effective assets and $\lambda = 10$ maintains 6–8. Overall, the bootstrap distributions indicate that the qualitative ranking of MMR portfolios versus benchmark strategies is preserved across a wide range of resampled paths, suggesting that the documented advantages are not driven by a single, lucky realization of market history.

6. CONCLUSION

The empirical and theoretical analysis confirms that applying Mean Absolute Deviation to regret rather than to returns provides a coherent, behaviorally grounded alternative to classical risk-based portfolio optimization. In the multi-asset ETF universe considered, regret-based portfolios deliver higher long-horizon returns and competitive risk-adjusted performance relative to standard allocation rules, but these gains are sample- and universe-specific and should be viewed as illustrative rather than universal. The framework explicitly targets opportunity cost—defined as the shortfall versus the ex-post best-performing asset—rather than absolute volatility, and introduces a single parameter

λ to govern the trade-off between low average regret and stable benchmark-relative performance.

Principal findings. The empirical analysis on fourteen liquid ETFs over June 2010–December 2025 highlights four main results. First, low but positive dispersion penalties ($\lambda = 0.1$) achieve the highest observed risk-adjusted performance, with a Sharpe ratio of 0.92 and CAGR of 14.82%, exceeding the Maximum Sharpe portfolio on both metrics (0.83 and 9.55%, respectively). This suggests that regret-based winner identification provides a more robust return signal than static first-moment estimates in this sample. Second, regret portfolios systematically outperform risk-based allocations in absolute returns: the $\lambda = 0.1$ specification exceeds Risk Parity (3.00%) and Inverse Volatility (2.95%) by approximately 12 percentage points in annualized return, while accepting higher volatility, underscoring that risk minimization and regret minimization target different objectives. Third, the framework exhibits a transparent regret–diversification trade-off: $\lambda = 0$ delivers the highest CAGR (16.10%) but is associated with extreme concentration and deep drawdowns (−37.63%), whereas $\lambda = 1$ produces materially improved diversification (5.4 effective assets) and still outperforms Equal Weight in CAGR (8.10% vs. 4.86%). Finally, as λ increases, portfolios converge not to equal weight but to allocations with roughly 7 effective assets, reflecting that only a subset of the universe persistently emerges as ex-post winners, while more defensive exposures rarely occupy the top rank.

Theoretical contributions. The study contributes on three fronts. Methodologically, it shows that applying a mean–MAD criterion to the distribution of regret yields a tractable objective that directly manages the volatility of opportunity cost while preserving the robustness properties of MAD-based optimization (Konno and Yamazaki, 1991). On the theoretical side, the paper connects the MMR criterion to the axiomatic regret literature (Savage, 1951; Sarver, 2008; Stoye, 2011) and establishes a formal nesting relationship with classical tracking-error optimization (Roll, 1992; Jorion, 2003). It also positions the framework relative to regret minimization in the online convex optimization tradition (Cover, 1991; Hazan, 2016). Behaviorally, it operationalizes regret preferences documented in experimental evidence (Bell, 1982; Loomes and Sugden, 1982; Bleichrodt and Wakker, 2015) by taking the regret series itself as the object of risk management, rather than adding a regret term to a conventional utility while leaving the risk measure unchanged (Baule et al., 2018).

Practical implications. For practitioners, the framework offers several advantages. First, regret provides a transparent performance metric for communication with boards and clients, making statements such as "average regret of 2.35% with MAD of 1.51%" more interpretable than volatility alone. Second, the approach internalizes the fact that asset managers are benchmarked against top performers, reducing the gap between how portfolios are constructed and how they are evaluated. Third, by relying on realized ex-post winners instead of explicit return forecasts, the framework sidesteps the primary weakness of mean–variance optimization—its dependence on noisy first-moment estimates (Chopra and Ziemba, 1993)—while remaining implementable with standard

tools. Finally, the single parameter λ provides explicit control over the regret–diversification trade-off, reducing reliance on multiple ad hoc constraints. In practice, different institutional mandates map naturally to different regions of the λ spectrum. Growth-oriented investors may find λ in the range $[0, 0.1]$ attractive, as these specifications combine strong long-run performance with acceptable risk. Balanced institutions (endowments, foundations) may gravitate toward $\lambda \approx 1$, while conservative mandates (pension funds, insurance portfolios) may prefer higher λ values that prioritize stability and breadth of exposure.

Limitations and further research. The asset universe is restricted to liquid U.S.-listed ETFs and excludes alternative assets, international fixed income and granular factor exposures; testing the framework in larger, more heterogeneous universes—especially those with liquidity and capacity constraints—is an important next step. The transaction-cost assumption of 30 basis points does not capture market impact, taxes or implementation shortfall, and the interaction between low turnover and costs warrants further analysis. The evaluation period, while spanning several major market episodes, remains finite; regime-dependent specifications of λ could enhance robustness. Although unreported robustness checks with top- K benchmarks confirm that the core trade-off is preserved, a systematic comparison across alternative benchmark definitions—including investable efficient-frontier portfolios and liability-driven references—merits further investigation, as does eliciting λ and preferred dispersion measures directly from investor choices, including tail-focused regret measures such as semi-variance or CVaR of regret. Extending the framework to multi-period settings with consumption, liabilities or stochastic investment opportunities would clarify how regret-based allocation interacts with the long-horizon objectives of pensions and endowments.

Notwithstanding these extensions, the core contribution stands: by shifting the risk object from portfolio returns to portfolio regret, the Mean–MAD Regret framework bridges the long-standing gap between how investors evaluate performance—relative to missed opportunities—and how portfolios are actually constructed. The combination of behavioral coherence, mathematical tractability, a single economically interpretable parameter, and demonstrated robustness across one thousand synthetic market histories makes the framework a practical and theoretically grounded addition to the institutional portfolio construction toolkit.

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